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School's in for Summer? The Effect of Encouraging Summer Community College Enrollment†

By Scott E. Carrell, Michal Kurlaender, Paco Martorell, and Christina Sun*

Enrollment in college courses during the summer can play an important role in supporting academic momentum, improving college persistence, and shortening time to degree (Brownback and Sadoff 2024). Yet fewer than one-third of college students enroll in summer coursework overall (Attewell and Jang 2013). For recent high school graduates, summer enrollment can be particularly consequential, offering an early opportunity to build academic engagement during the transition to college. This is especially salient for lower-income students, many of whom intend to enroll in college but ultimately do not. Despite a multitude of strategies to support students in their transition to college, and to increase degree completion, there is limited causal evidence on college course taking during the summer.

Summer enrollment at a community college immediately after high school could support students' transition to college through several potentially complementary channels. By keeping students connected to their postsecondary goals after graduation, it may help prevent summer melt (Castleman and Page 2014a, 2014b). Summer school enrollment can also sustain academic momentum by maintaining engagement with coursework and continuity in schooling (Attewell, Heil, and Reisel 2012). In addition, summer enrollment provides an early opportunity to complete developmental or gateway coursework, potentially accelerating time to degree. Relatedly, summer school enrollment following completion of high school could help students learn to navigate institutional systems such as registration and financial aid, building the "know-how" needed to overcome barriers that often derail students early in college, especially for disadvantaged students. Finally, taking summer courses can serve as a low-stakes way for students to "try on" college before starting their first term.

In this paper, we report results from a randomized experiment where we provide graduating high school students an information nudge about the benefits of summer school enrollment at a California Community College. The nudge included information about enrollment options, the potential for tuition aid, and the transferability of classes to a four-year college. The experiment was conducted on a population of approximately 8,000 California high school seniors who applied for financial aid by submitting the Free Application for Federal Student Aid (FAFSA) or California Dream Act Application (CADAA) for the 2023–2024 academic year, and who participated in a survey about their intended college plans. We matched these high school seniors to their college enrollment information to determine whether treated students were more likely to enroll in summer school. We find that this light-touch informational nudge results in a 17 percent increase in the likelihood of community college enrollment during the summer and a 23 percent increase in community college units earned during the summer. These results show that low-cost, information-based interventions can meaningfully shift early postsecondary trajectories and offer a promising lever for supporting students' transition into college.

*Carrell: University of Texas at Austin (email: scott.carrell@austin.utexas.edu); Kurlaender: University of California, Davis (email: mkurlaender@ucdavis.edu); Martorell: University of California, Davis (email: pmartorell@ucdavis.edu); Sun: University of California, Davis (email: ucsun@ucdavis.edu). This study was registered in the AEA RCT Registry, ID AEARCTR-0011439 (Carrell et al. 2023). UC Davis IRB Approval ID 2041454–1. Thank you to Rachel Baker, Brooks Bowen, Ben Castleman, Michael Hill, Di Xu, and the other conference participants for your helpful comments and suggestions.
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I. Research Design

We conducted a randomized field experiment to examine whether providing timely information about summer college enrollment opportunities encourages high school seniors who are about to graduate to enroll in summer coursework at their local community college. The study population consisted of California high school students who completed a survey about their postsecondary plans in the spring of their senior year, and who indicated that they planned to enroll in college in the upcoming fall term.

Survey respondents were assigned to either the treatment or control condition using simple random assignment. All students (i.e., both treated and control students) were asked about whether they planned to take college classes over the summer. Students in the treatment group received a brief message highlighting the benefits of enrolling in summer courses at their local community college. The message read:

> *Did you know that taking college classes this summer could help you finish college faster and save money along the way?* At your local community college:

- There are many in-person and online options.
- Credits from most classes can be transferred to a UC/CSU.
- Many students qualify for free tuition.

After respondents read this message, we asked, "*After taking this information into account, how likely are you to take college classes this summer?*", with response options on a five-point Likert scale ("Definitely not" … "Definitely will").

The control group did not receive any message. The intervention was intentionally light touch, designed to mimic scalable modes of communication commonly used by school districts and colleges to support academic momentum and other positive enrollment behaviors. Random assignment occurred at the individual level and was implemented using the survey administration platform, ensuring allocation concealment and minimizing the risk of spillover across students.

II. Data

The survey was administered in May 2023 to all California high school seniors who submitted a FAFSA or CADAA for the 2023–2024 academic year. As a result, this college-bound population is more likely to be low income, as compared to the full college-bound population of California high school seniors. The survey response rate is low at 4 percent but is representative of the population of FAFSA submitters (Hurtt et al. 2024).

The analytic sample consists of 8,222 observations from the survey (4,110 in the treatment group and 4,112 in control). The survey data also include a host of demographic information (e.g., race, gender, high school type (public/private), parent education, and whether English is the primary language spoken in the home).We merge the survey to two administrative data files. First, the California Student Aid Commission (CSAC), the state agency that administers financial aid programs. These data include derived income ranges and expected family contribution ranges. Finally, to measure enrollment outcomes, we merge the data to information provided by the California Community College Chancellor's Office, specifically whether students enrolled in the summer following high school completion, units enrolled and earned in summer, and performance (i.e., grade point average in summer courses).

Table 1 includes summary statistics of the population of California seniors in 2023–2024 receiving the survey and survey respondents (by experimental condition). These variables include information collected from the survey as well as information from CSAC records available for all students sent the survey (including survey nonrespondents). Compared to the full sample of financial aid applicants, average income among survey respondents is about 6 percent higher, and average high school GPA is about 2.2 percent higher. Sample means of baseline variables among survey respondents, including

TABLE 1—SUMMARY STATISTICS

Variable	FAFSA-submitting cohort	Control	Treatment
Black		0.034	0.042
Asian/Filipino		0.142	0.138
Hispanic		0.432	0.426
White		0.202	0.203
Two or more races		0.166	0.169
Woman		0.581	0.566
Man		0.353	0.369
Public high school		0.814	0.820
First-gen. college student		0.378	0.391
Baseline intention: no		0.565	0.568
Baseline intention: not sure		0.238	0.231
Baseline intention: yes		0.196	0.200
High school GPA	3.21	3.267	3.293
Derived income (thousands)	124.435	131.867	130.071
Expected family contribution (thousands)	27.080	32.641	31.600
Observations	286,163	4,114	4,112

Notes: Not shown here are other races and Native American. Gender is self-reported and coded as mutually exclusive categories; public high school includes public charter schools. First-generation college student is a binary variable based on parent education level (never attended college).

TABLE 2—IMPACTS ON COMMUNITY COLLEGE SUMMER SCHOOL ENROLLMENT

	Enrolled in community college summer school (1)	Units attempted (2)	Units earned (3)
Summer nudge treatment	0.012	0.055	0.062
	(0.006)	(0.030)	(0.027)
Control group outcome mean	0.063	0.273	0.232

Notes: Table entries are the mean difference in a given outcome by treatment status with robust standard errors in parentheses. The sample size is 8,226 for all outcomes.

stated intentions about summer school enrollment, are similar for treated and control students (note that these variables are not available for students who did not complete the survey). Differences by experimental condition are not statistically significant (F-statistic: 0.86).

III. Results

We first find a positive, statistically significant, effect on treated students' reported likelihood of enrolling in summer school. Specifically, treated students were 17.6 percentage points more likely to indicate that they planned to take community college summer courses after reading the information nudge than they were before (19.8 percent compared to 37.4 percent; $p < 0.001$). This provides evidence that students in the treatment group received and understood the information, and that, for some students at least, it changed their stated intentions about taking summer school courses.

The primary outcome of interest is enrollment in a community college summer course measured using postsecondary administrative records. Table 2 presents treatment-control differences for outcomes: any enrollment, units enrolled, and units earned. The summer school nudge increased summer enrollment the number of summer units, including summer units earned. Specifically, the nudge induced enrollment for summer school by 1.2 percentage points (nearly 17 percent of the control group mean), and by 0.062 more units earned (26.7 percent of the control group mean). We find no

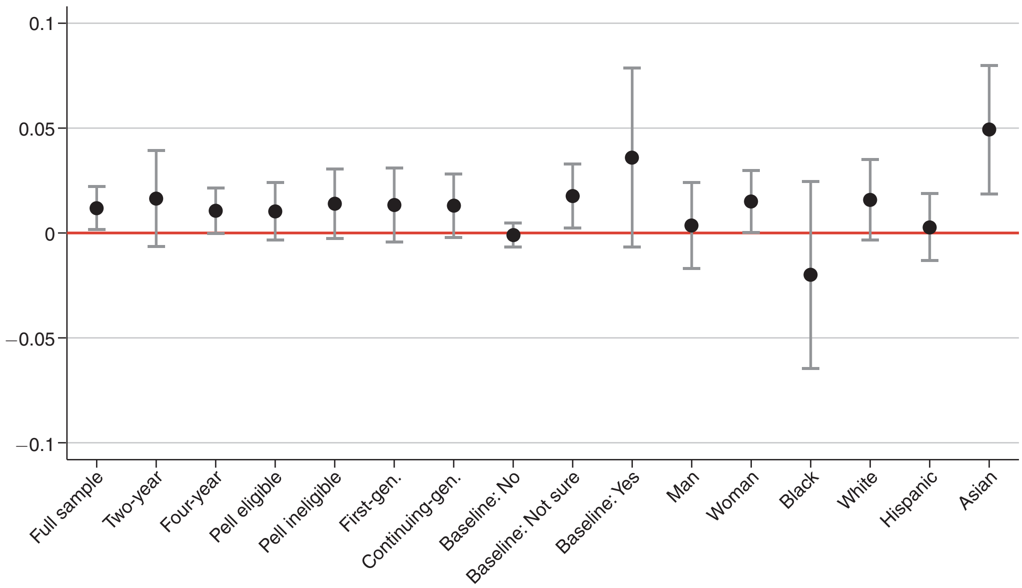

Figure 1. Heterogeneity in the Effect of the Nudge on Summer School Enrollment

Note: Figure shows estimated impacts of the nudge on the likelihood of enrolling in community college summer courses for different student subgroups; bands represent 95 percent confidence intervals.

statistically significant relationship for grade point average (results not reported). Estimates from regression models that include controls (race, gender, high school sector, language spoken at home, parental education, parent income, high school GPA, expected family contribution) are very similar in magnitude and statistical significance.

We also examined whether the treatment effects varied by student characteristics, and Figure 1 shows treatment effect estimates on the likelihood of enrolling in summer school for different subgroups. Although we have limited statistical power, which prevents us from drawing definitive conclusions, there are several interesting suggestive patterns. First, the effects do not appear to differ much between whether students plan to attend a two- or four-year college, or whether they are first-generation students. On the other hand, treatment effects appear to increase in baseline stated likelihood of taking summer school classes, and are larger for women and White and especially Asian students.

IV. Conclusion

Early enrollment in summer coursework can play an important role in supporting students' transition from high school to college. Our findings suggest that this low-touch information nudge can induce the short-run outcomes of summer school enrollment. By enrolling in summer courses at a local community college, recent high school graduates can maintain a continuity in learning that could improve persistence and degree attainment, reduce summer melt, and provide opportunities to earn transferable credits before the fall semester, reducing future course loads and helping students make progress toward degree requirements. In this context, the increases in summer enrollment generated by our light-touch informational intervention are noteworthy; even a modest prompt delivered at a critical moment can activate mechanisms known to promote academic momentum and degree progress. Taken together, the experimental findings and prior research suggest that encouraging summer coursework represents a scalable strategy for strengthening students' transitions into postsecondary education.

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